

Real Estate Price & Feature Exploration Using SQL Server

NTISQL Project

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Academic Insights

Data-Driven Perspectives
for Scholarly Excellence

PRESENTATION
2024



Project Overview

Dataset

Ames Housing Dataset

Records

Number of properties: 1,460

Main Target

SalePrice

Tool

Microsoft SQL Server

Goal

Explore factors associated with house prices using SQL

Dataset & Main Columns

Key columns used in analysis (concise):

- Id
- SalePrice
- OverallQual
- Neighborhood
- YearBuilt
- GarageCars
- GrLivArea
- BedroomAbvGr
- TotalBsmtSF
- Heating

DATASET COLUMNS

COLUMN NAME	DATA TYPE	DESCRIPTION	EXAMPLE	REQUIRED
user_id	INTEGER	Unique identifier for the user	1024	✓
user_name	STRING	Full name of the user	Ava Smith	✓
email	STRING	Email address of the user	ava.smith@example.com	✓
signup_date	DATE	Date when the user signed up	2024-04-21	✓
plan_type	STRING	Subscription plan type	Premium	✓
is_active	BOOLEAN	Indicates if the user is active	true	✓
last_login	DATETIME	Timestamp of the last login	2024-05-19 08:15:33	✓
login_count	INTEGER	Total number of logins	27	✓
country	STRING	Country of residence	United States	✓
referral_code	STRING	Referral code used (if any)	REF-8X2Q	✓

Query 1 — Total Houses & Average Sale Price

Question

What is the total number of houses and the average sale price?

SQL

```
SELECT COUNT(Id) AS Total_Houses, AVG(SalePrice) AS Avg_Sale_Price FROM train;
```

Output

Result table: (actual output from the database — leave space for pasted results)

Total_Houses	Avg_Sale_Price
1460	180921

Query 2 — Top 5 Most Expensive Houses

Question

What are the top 5 most expensive houses?

SQL

```
SELECT TOP 5 Id, Neighborhood, YearBuilt, SalePrice FROM train ORDER BY SalePrice DESC;
```

Output

Top 5 rows returned by the query — include exact rows from the database:

Id	Neighborhood	YearBuilt	SalePrice
692	NoRidge	1994	755000
1183	NoRidge	1996	745000
1170	NoRidge	1995	625000
899	NridgHt	2009	611657
804	NridgHt	2008	582933

Query 3 — Average Price by Overall Quality

Question

What is the average house price for each Overall Quality rating?

SQL

```
SELECT OverallQual, AVG(SalePrice) AS Avg_Price, COUNT(*) AS House_Count FROM train GROUP BY OverallQual ORDER BY OverallQual DESC;
```

Output

Actual query output — include full table as returned by SQL Server:

OverallQual	Avg_Price	House_Count
10	438588	18
9	367513	43
8	274735	168
7	207716	319
6	161603	374
5	133523	397
4	108420	116
3	87473	20
2	51770	3
1	50150	2

Query 4 — Top 5 Neighborhoods by Number of Houses

Question

What are the top 5 neighborhoods by number of houses?

SQL

```
SELECT TOP 5 Neighborhood, COUNT(*) AS Total_Houses, AVG(SalePrice) AS Avg_Price FROM train GROUP BY Neighborhood ORDER BY Total_Houses DESC;
```

Output

Actual returned rows — include exact results:

Neighborhood	Total_Houses	Avg_Price
NAmes	225	145847
CollgCr	150	197965
OldTown	113	128225
Edwards	100	128219
Somerst	86	225379

Query 5 — Garage Capacity vs Average Price

Question

How does garage capacity affect average house price?

SQL

```
SELECT GarageCars, AVG(SalePrice) AS Avg_Price, COUNT(*) AS Total_Houses FROM train GROUP BY GarageCars ORDER BY GarageCars;
```

Output

Actual results table (and chart if available from real output):

GarageCars	Avg_Price	Total_Houses
0	103317	81
1	128116	369
2	183851	824
3	309636	181
4	192655	5

Query 6 — New vs Old Houses

Question

How do modern houses compare with older houses in terms of average price?

SQL

```
SELECT CASE WHEN YearBuilt >= 2000 THEN 'New (>=2000)' ELSE 'Old (<2000)' END AS House_Age_Category, AVG(SalePrice)
AS Avg_Price, COUNT(*) AS Total_Houses FROM train GROUP BY CASE WHEN YearBuilt >= 2000 THEN 'New (>=2000)' ELSE
'Old (<2000)' END;
```

Output

Actual query output — include exact rows. Use a two-bar comparison if actual values are present.

House_Age_Category	Avg_Price	Total_Houses
New (>=2000)	242439	388
Old (<2000)	158655	1072

Query 7 & 8 — Large Houses Count & Avg Price by Bedrooms

Query7—Question

How many houses have a living area greater than 2,000 square feet?

SQL

```
SELECT COUNT(*) AS Large_Houses_Count FROM train WHERE GrLivArea > 2000;
```

Output (Query 7)

Large_Houses_Count
214

Query 8 — Question

What is the average house price according to the number of bedrooms?

SQL

```
SELECT BedroomAbvGr, AVG(SalePrice) AS Avg_Price FROM train GROUP BY BedroomAbvGr ORDER BY BedroomAbvGr;
```

Output (Query 8)

BedroomAbvGr	Avg_Price
0	221493
1	173162
2	158197
3	181056
4	220421
5	180819
6	143779
8	200000

Query 9 — Price extremes and range

Question: What is the minimum price, maximum price, and price range?

SQL

```
SELECT MAX(SalePrice) AS Max_Price, MIN(SalePrice) AS Min_Price, (MAX(SalePrice) - MIN(SalePrice)) AS Price_Range FROM train;
```

Max_Price	Min_Price	Price_Range
75500	34900	720100

Query 10 — Houses with SalePrice > \$400,000 (Luxury outliers)

SQL

```
SELECT Id, Neighborhood, GrLivArea, SalePrice FROM train WHERE SalePrice > 400000 ORDER BY SalePrice DESC;
```

Question: Which houses can be considered luxury outliers with a sale price above \$400,000?

Id	Neighborhood	GrLivArea	SalePrice
692	NoRidge	4316	755000
1183	NoRidge	4476	745000
1170	NoRidge	3627	625000
899	NridgHt	2364	611657
804	NridgHt	2822	582933
1047	StoneBr	2868	556581
441	NridgHt	2402	555000
770	StoneBr	3279	538000
179	StoneBr	2234	501837
799	NridgHt	3140	485000

Query 11 — Average total house area

Question: What is the average total house area, combining living area and basement area?

SQL

```
SELECT AVG(GrLivArea + TotalBsmtSF) AS Avg_Total_House_Area FROM train;
```

Avg_Total_House_Area
2572

Query 12 — Distribution and average price by heating type

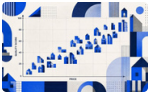
Question: How are houses distributed by heating type and what is the average price for each heating type?

SQL

```
SELECT Heating, COUNT(*) AS Total_Houses, AVG(SalePrice) AS Avg_Price FROM train GROUP BY Heating ORDER BY Total_Houses DESC;
```

Heating	Total_Houses	Avg_Price
GasA	1428	182021
GasW	18	166632
Grav	7	75271
Wall	4	92100
OthW	2	125750
Floor	1	72500

Key findings from the exploratory analysis



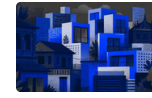
OverallQual → SalePrice

OverallQual shows a strong positive relationship with SalePrice: higher quality ratings consistently map to higher sale prices.



High-quality premium segment

Houses with OverallQual 8–10 have average prices noticeably higher than those rated 5–6, signaling clear market segmentation by quality.



YearBuilt impact

Homes built in 2000 or later show higher average sale prices compared with older houses—reflecting newer construction premiums.



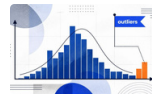
Living area premium

GrLivArea above ~2,000 sq ft is associated with materially higher market value — living space drives pricing power.



Garage capacity

GarageCars capacity of 2–3 correlates with higher house values, suggesting practical amenities add measurable price lift.



Outliers identified

Homes with SalePrice > \$400,000 were flagged as potential outliers for targeted review and treatment in modeling.

Machine learning readiness — features & preprocessing

Recommended features

- OverallQual
- GrLivArea
- GarageCars
- YearBuilt
- TotalBsmtSF

Suggested models

- Linear Regression —
baseline interpretability
- Random Forest —
robust to nonlinearity
and interactions

Preprocessing checklist

Missing values

Impute LotFrontage and other NA fields; consider indicator flags when missingness is informative.

Outlier handling

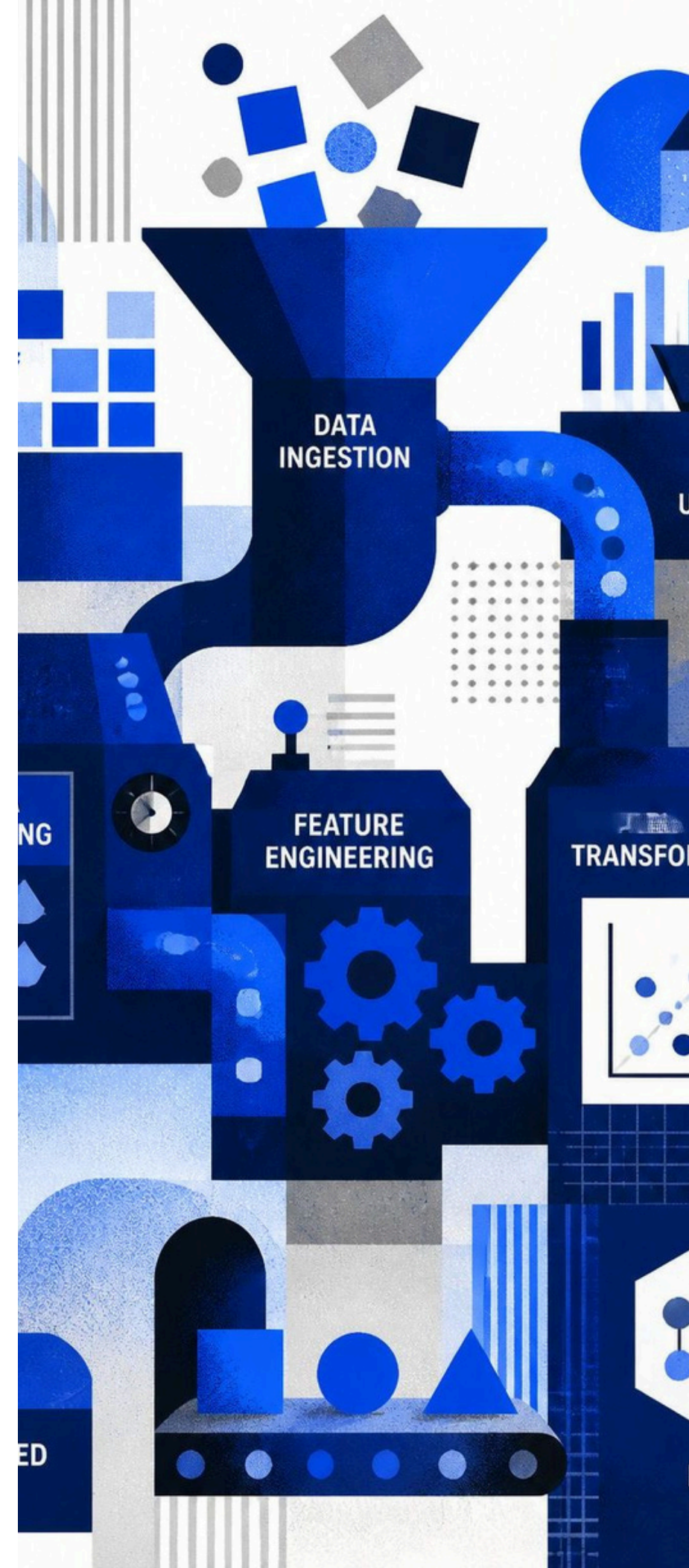
Identify extreme SalePrice and GrLivArea values; consider winsorizing or separate treatment for modeling.

Feature engineering

Create combined area features, age (YrSold - YearBuilt), and categorical encodings for neighborhoods.

Scaling & validation

Scale numeric features as needed and use stratified CV where price distribution is skewed.



Conclusion — project summary & next steps

The exploratory SQL analysis clarified key price drivers and prepared the dataset for modeling:

- Quality (OverallQual), living area (GrLivArea), basement area, garage capacity, and YearBuilt emerged as primary predictors of SalePrice.
- Houses above \$400K were identified as potential outliers and should be reviewed before training models.
- Preprocessing must address missing LotFrontage values, extreme outliers, and appropriate feature engineering (combined area, age, neighborhood encodings).
- Recommended next steps: finalize preprocessing pipeline, run baseline Linear Regression, then evaluate Random Forest for improved accuracy and robustness.

Thank You!

Thank you for your attention.

Presented by:

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NTI SQL Project

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